

3.12 understand how the developing embryo is protected by amniotic fluid

3.13 recall the roles of oestrogen and testosterone in the development of secondary sexual characteristics.

b) Inheritance

- 3.14 recall that the nucleus of a cell contains chromosomes on which genes are located
- 3.15 understand that a gene is a section of a molecule of DNA
- 3.16 describe a DNA molecule as two strands coiled to form a double helix, the strands being linked by a series of paired bases: adenine (A) with thymine (T), and cytosine (C) with guanine (G)
- 3.17 understand that genes exist in alternative forms called alleles which give rise to differences in inherited characteristics
- 3.18 recall the meaning of the terms: dominant, recessive, homozygous, heterozygous, phenotype, genotype and **codominance**
- 3.19 describe patterns of monohybrid inheritance using a genetic diagram
- 3.20 understand how to interpret family pedigrees
- 3.21 predict probabilities of outcomes from monohybrid crosses
- 3.22 recall that the sex of a person is controlled by one pair of chromosomes, XX in a female and XY in a male
- 3.23 describe the determination of the sex of offspring at fertilisation, using a genetic diagram
- 3.24 understand that division of a diploid cell by mitosis produces two cells which contain identical sets of chromosomes
- 3.25 understand that mitosis occurs during growth, repair, cloning and asexual reproduction
- 3.26 understand that division of a cell by meiosis produces four cells, each with half the number of chromosomes, and that this results in the formation of genetically different haploid gametes
- 3.27 understand that random fertilisation produces genetic variation of offspring
- 3.28 recall that in human cells the diploid number of chromosomes is 46 and the haploid number is 23
- 3.29 understand that variation within a species can be genetic, environmental, or a combination of both
- 3.30 recall that mutation is a rare, random change in genetic material that can be inherited
- 3.31 describe the process of evolution by means of natural selection
- 3.32 understand that many mutations are harmful but some are neutral and a few are beneficial
- 3.33 understand how resistance to antibiotics can increase in bacterial populations